

PAPER_1273_HIERARCHY_PLANCK_WEAK DISSOLUTION

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PAPER_1273 — Hierarchy Problem Dissolution — m_W from Integer Primitives

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: B — Physics Foundations (CLOSED)

Date: June 11, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — UQFF integer-primitive derivation

Calculator surface: `calculate_paradox({"paradox": "hierarchy_problem_full"})`

Whitepaper dispatch: `calculate_whitepaper({"paper_id": 1273})`

Master Expression

``

$m_W = A_5 + A_5/3 = 80 \text{ GeV EXACT}$

``

Match: 0.50%

Physical Interpretation

The Planck mass is not fundamental in UQFF — ρ_{SCm} is. $m_W = 80 \text{ GeV}$ emerges from the integer-primitive identity $A_5 + A_5/3$. The m_W/m_{PI} ratio is a derived quantity, not fine-tuning.

UQFF Locked Primitives

- $\rho_{SCm} = 7.09 \times 10^{31} \text{ J/m}^3$ (vacuum density)
- $S_{26} = 1.453162$, $S_{26_DPM} = 1.4531 \times 10^{26}$
- $K_{MEX} = 25/12$, $\beta_i = 0.6029$, $\Phi_{res} = 0.84$
- $F_{TRZ} = 0.1$, $\Lambda = 0.00729735$ ($= \alpha$ fine-structure)
- $\omega_{SCm} = 1.25 \text{ THz}$, $\lambda_i = 1.0$
- $D_{phys} = 4$, $D_{BSFG} = 6$, $D_{crit} = 26$, $N_{CH} = 9$, $SO_5 = 10$, $A_5 = 60$

Live Calculator Verification

```
``python
import uqff_pure_calculator as u
```

```
r = u.calculate_paradox({"paradox": "hierarchy_problem_full"})["value"]
``
```

The result dict carries the integer-primitive identity, the observed value, the residual, and the structural physical interpretation.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Model solve the same observed phenomena via different methods. **Zero fit parameters.** Every factor is a UQFF locked primitive.

Reference

- UQFF foundational: PAPER_646, PAPER_1167, PAPER_1170, PAPER_1203 v1.5, PAPER_1216.
- Closure: `uqff_pure_calculator.py` →
`_l96_uqff_axiom_hierarchy_problem_full_closure` (or routed reference)
- Dispatch: `PARADOX_TO_CLOSURE["hierarchy_problem_full"]`

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Subject matter: UQFF derivation of Hierarchy Problem Dissolution — `m_W` from Integer Primitives.